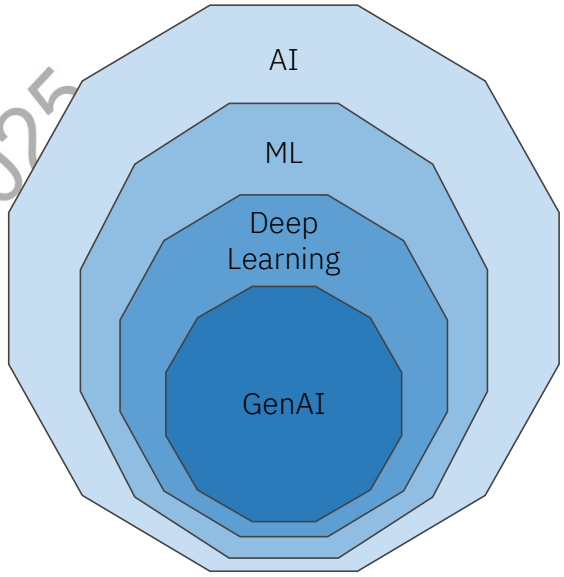
Introduction to AI

Week 4: Statistical learning

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090 830 6668
Data & AI Leader, IBM ASEAN

Tran Dang An
Assistant

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Objectives

- Understand the idea of *statistical learning*
- Explain and use Naive Bayes, Decision Tree
- Case study: Spam Email Classification

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AI Hierarchy

Artificial Intelligence

W1: Basic concepts, history of AI
W2: Knowledge representation and logic
W3: Searching algorithms (pathfinding and text)

Machine Learning

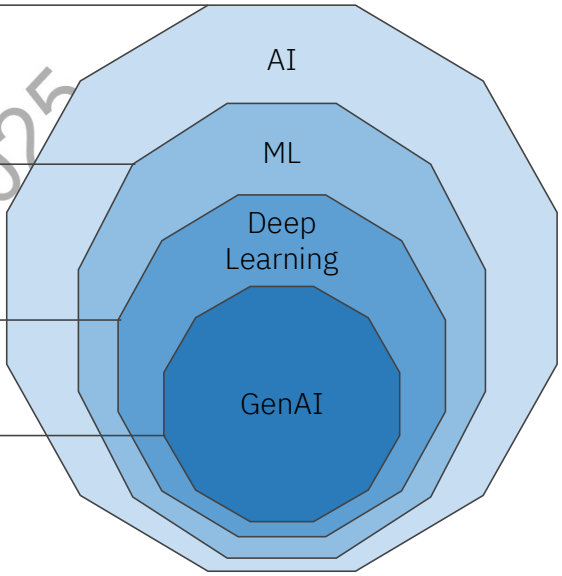
W4: Statistical learning
W5: Supervised machine learning
W6: Unsupervised machine learning

Deep Learning

W7: Perceptron learning and neural network learning
W8: Deep learning

Generative AI

W9 + W10: Modern AI and development in real world applications (LLM & GenAI)



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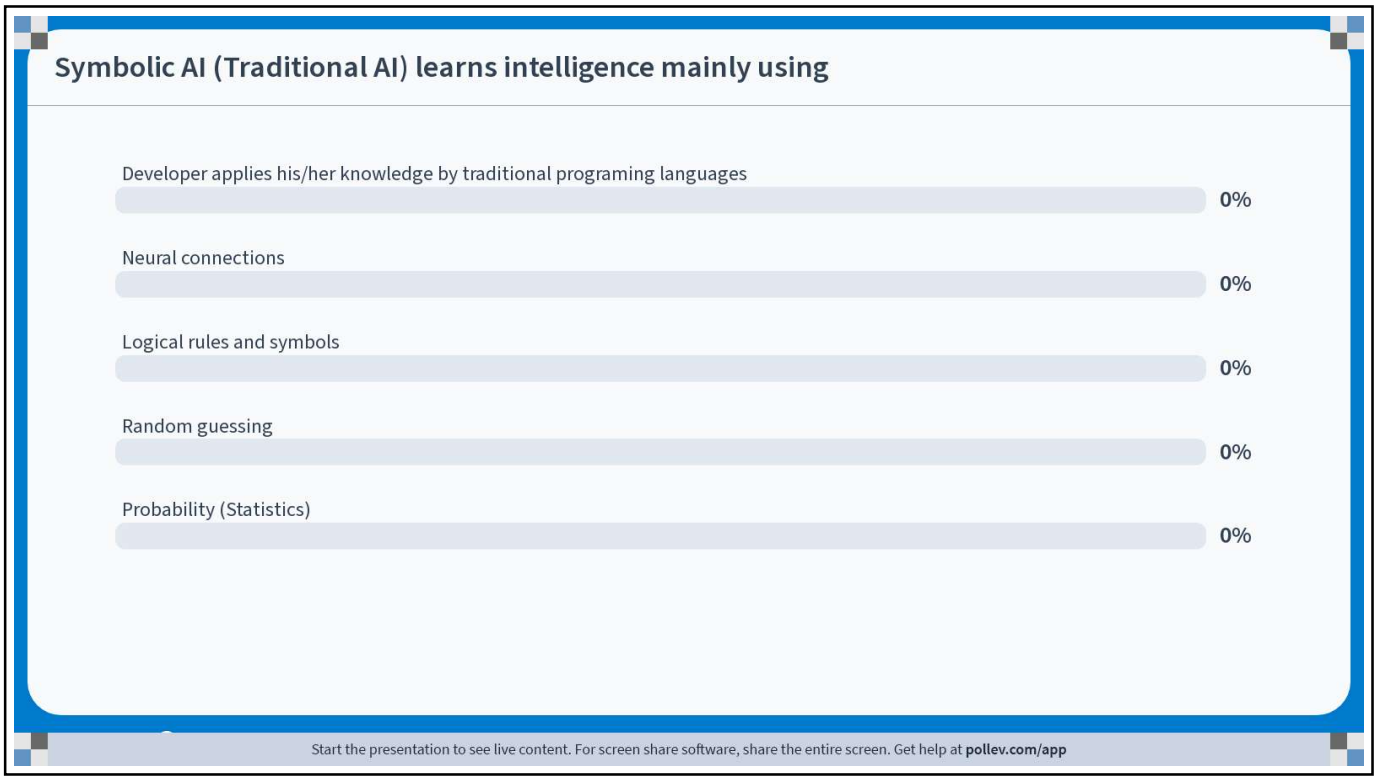
Contents

- Statistical Learning Overview
  - Naive Bayes
  - Decision Trees

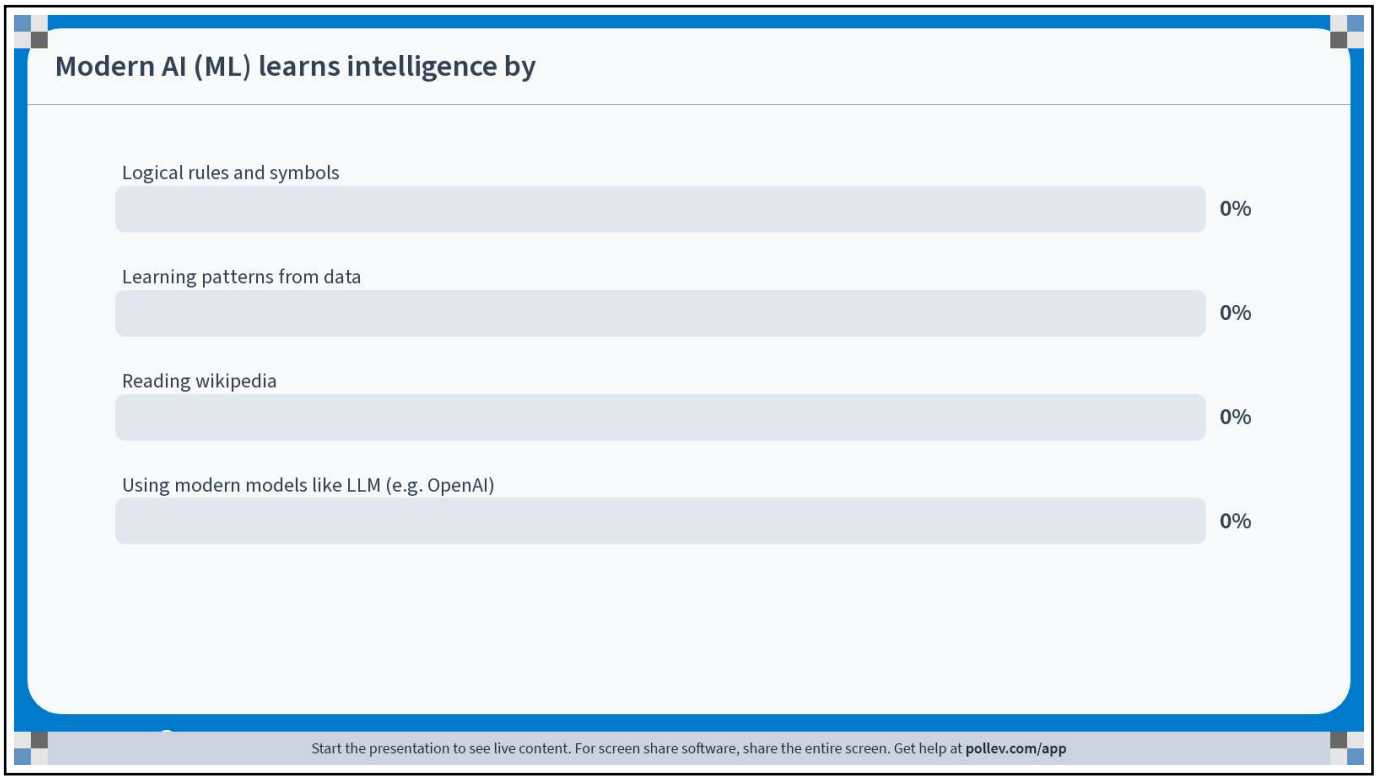
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Symbolic AI (Traditional AI) learns intelligence mainly using
Developer applies his/her knowledge by traditional programing languages
0%
Neural connections
0%
Logical rules and symbols
0%
Random guessing
0%
Probability (Statistics)
0%
Start the presentation to see live content. For screen share software, share the entire screen. Get help at pollev.com/app

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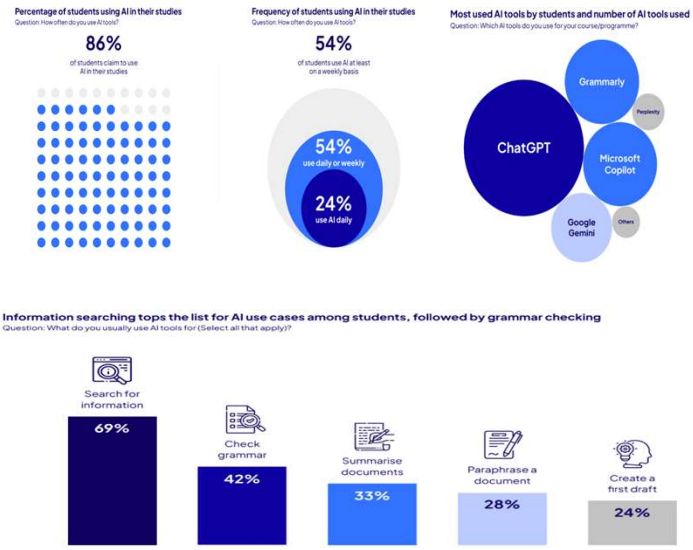

Modern AI (ML) learns intelligence by
Logical rules and symbols
0%
Learning patterns from data
0%
Reading wikipedia
0%
Using modern models like LLM (e.g. OpenAI)
0%
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What is Statistics?

Statistics is the science of collecting, analyzing, and interpreting data. It helps us describe patterns and make informed decisions

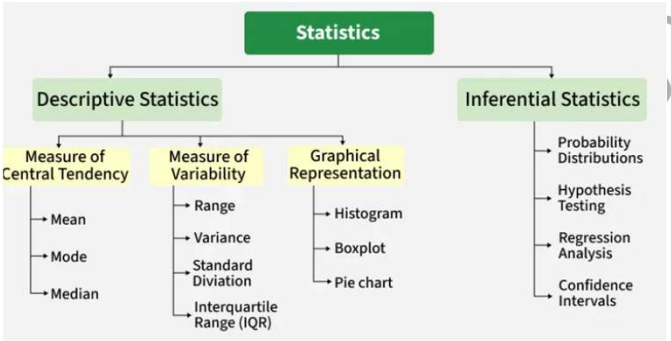
Statistics turns **raw data → information → insight**

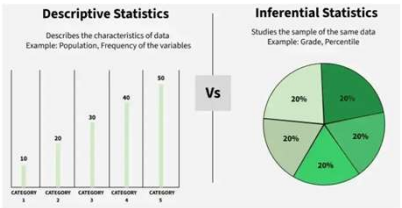


Source: Digital Education Council AI Student Survey, 2024

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Statistics for Data Science





https://www.geeksforgeeks.org/data-science/statistics-for-data-science/

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From Symbolic/Traditional AI to Statistical Learning

🧠 Symbolic/Traditional AI

- **Rule-based & symbolic**: logic, reasoning, explicit knowledge
- **Human-designed rules**: effective for structured, well-defined tasks
- **Limitations**: brittle in messy data; cannot learn or adapt automatically

📊 Statistical Learning

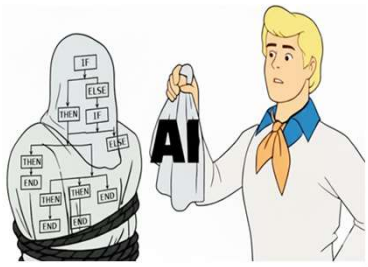
- **Shift to data-driven learning** instead of hand-crafted rules
- **Models learn patterns** between inputs (X) and outputs (Y) using statistics
- **Strengths**: handles noisy data, real variation, and predictive tasks

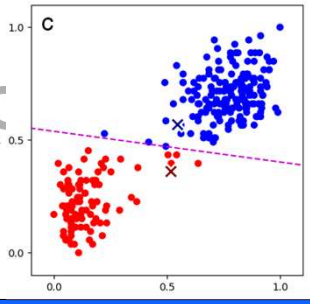
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Why This Transition Matters





Symbolic/Traditional AI	Statistical Learning
Symbolic logic	Data-driven patterns
Deterministic rules	Probabilistic models
Human knowledge → machine	Machine discovers knowledge
Harder to scale	Scales with more data

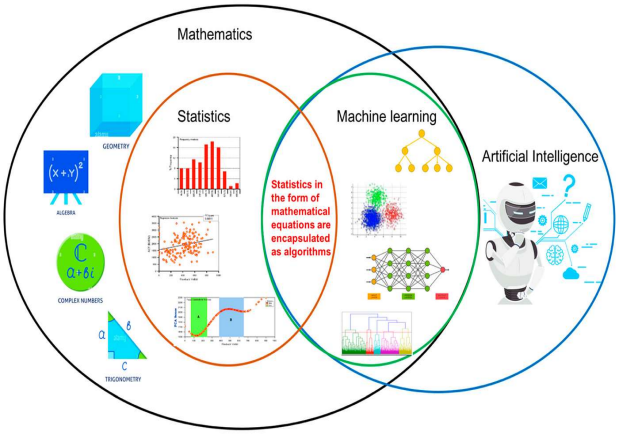
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Statistical Learning is the Foundation

- ML is fundamentally about learning patterns from data — exactly what statistical learning defines.
- Key ML ideas: parameters, uncertainty, prediction, generalization, loss, inference, … all come from statistics.
- Statistical thinking explains why models work, when they fail, and how to improve them, without it, models are black boxes.



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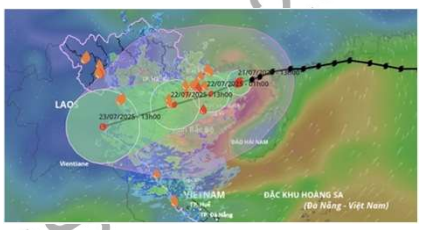
Why Statistics Matters in Real Life

We constantly make decisions under uncertainty, and statistics guides better choices

Simple relatable examples:



Traditional Weather Forecast: Before satellites, Vietnamese elders relied on a premium forecasting system, zero technology, 100% rural AI.



Modern Weather Forecast: Forecasting rain uses statistical probability



Shopping: Netflix / Spotify/ Shopee recommendations

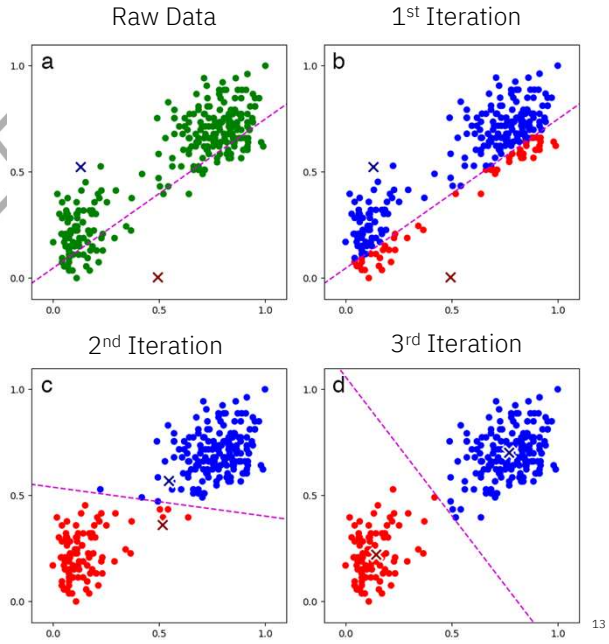
Statistics matters because **data alone isn't useful — understanding it is.**

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Learning from Data by statistical method (1)

- Real life data is messy: people behave differently, data is noisy, and patterns aren't obvious.
- We rarely see clean, perfect rules in reality.
- We need statistical learning to summarize the world, uncover hidden relationships, and make reliable predictions.



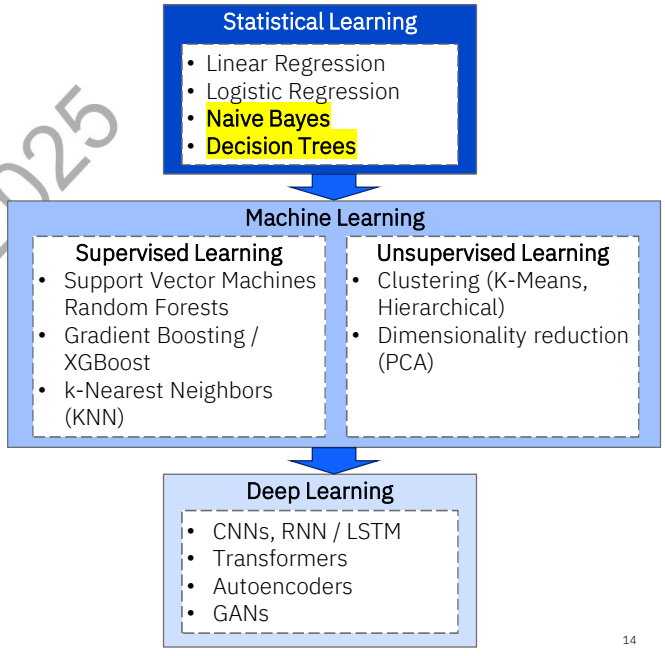
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Statistical Learning in AI

- Once patterns are learned, models help us forecast things like spam, price, or demand.
  - More data improves decisions — models evolve and refine boundaries over time.
- AI evolved from early statistical learning where models estimated probabilities to modern AI (ML), where systems learn patterns from data and improve automatically.



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Definition of Statistical Learning

A set of tools for:

- **Understanding the relationship** between **inputs** (X) and **output** (Y)
- **Predicting** the output for **new, unseen** inputs
- **Quantifying uncertainty** in predictions

Formally:

We want to estimate this function from data

$$f(X) = Y$$

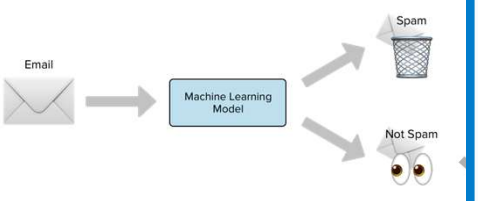
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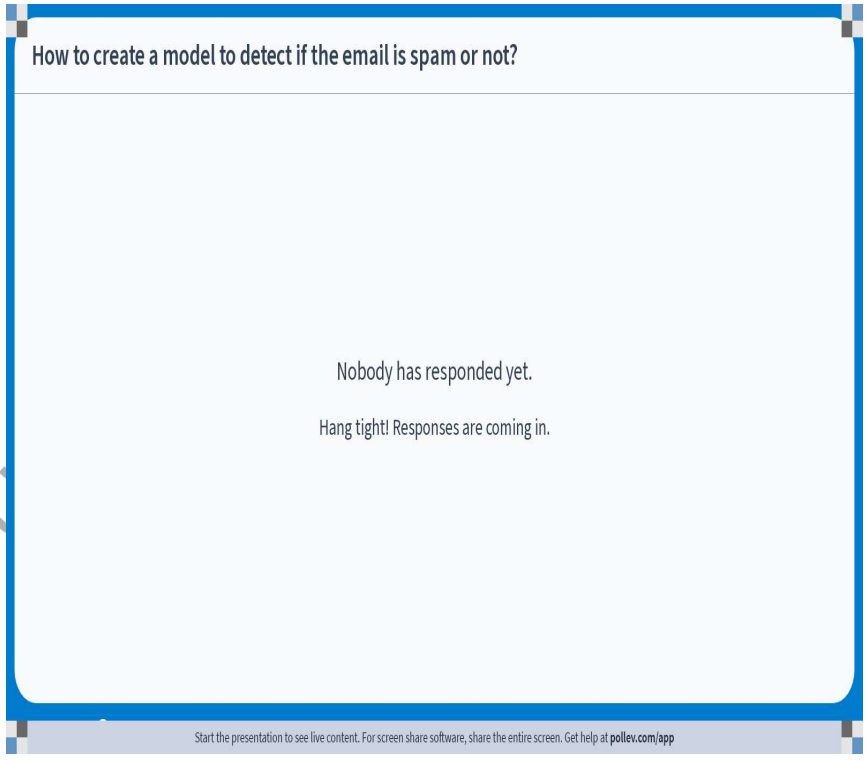
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Case study: Spam Email Classification

Model: Spam Email Classification





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Case Study: Spam Email Classification

Prediction

Use features $\boldsymbol{X}$ → Predict $\boldsymbol{Y}$

Inference

Understand how each feature influences the outcome

Problem Statement

Predict whether an email (Y) is **Spam** or **Not Spam**

Selected features X:
- Contains spam words (free, winner)
- Sender type (known, unknown)
- Email length (short, medium, long)

Inference
- Are emails with more spam words more likely to be spam?
- How strong is that effect to the prediction

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- Decision Trees

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Bayes' Theorem (Formal)

Definition:

$$P(A|B) = P(A) \times \frac{P(B|A)}{P(B)}$$

posterior = $P(A|B)$; prior = $P(A)$; likelihood = $P(B|A)$; marginal = $P(B)$

Where:

- $\boldsymbol{Y}$: class label (Spam / Not Spam)
- $\boldsymbol{X}$: observed features
- $\boldsymbol{P(A)}$ prior probability
- $\boldsymbol{P(A \mid B)}$: posterior probability (what we want)
- $\boldsymbol{P(B \mid A)}$: likelihood

Notes:

- *P(Y): Probability of Y (xác xuất mục tiêu)*
- *P(Y|X): Probability of Y given X (xác xuất mục tiêu với điều kiện đầu vào X)*
- *P(X|Y): Likelihood (Khả năng của X khi biết mục tiêu Y)*

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Naive Bayes Approach

Many AI tasks rely on estimating how likely something is, given evidence
Example, instead of asking

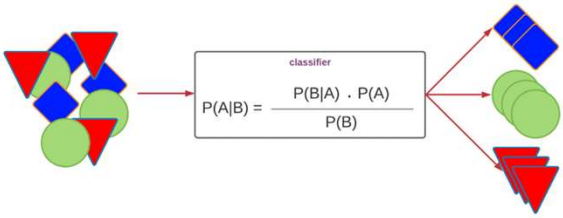
If it contains the word "free", how likely is this email to be spam?
-> given X, what is P(Y): P(Y|X) *Probability*

Naïve Bayes asks:

How often do we see these features in spam or no-spam emails?
-> given Y, what is P(X): P(X|Y) *Likelihood*

Common use cases:

- Spam detection
- Document classification
- And more ...



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Example

- **Probability (xác suất)**: xác suất xảy ra của dữ kiện khi biết mô hình
  P(data | model)
- **Likelihood (hàm khả năng)**: mức độ hợp lý của mô hình/tham số khi biết dữ kiện
  L(model | data)

- Let's take this example
  - X = "It is raining,"
  - Y = "The ground is wet."
- Then P(Y|X) is (*Probability*)
  - The probability the ground is wet **given that** it is raining
- What is about P(X|Y) (*Likelihood*)
  - Given the fact that ground is wet, what is the likelihood of raining?
- What are about Prior Probability P(Y) and Posterior P(Y|X)?
  - P(Y): based on the actual training data, 40% the ground is wet (*prior before given data input*)
  - P(Y|X): given data input X (e.g raining), we can predict the P of Y given X more accurate, e.g. 80% ground is wet (*posterior after given data*)

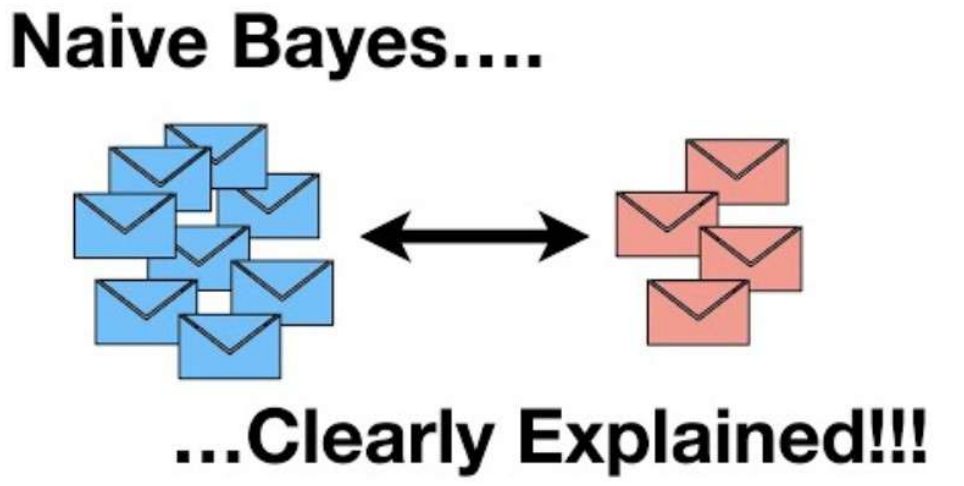
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Terms
✓*Prior probability = xác suất tiên nghiệm*
✓*Posterior probability = xác suất hậu nghiệm*

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Naive Bayes Classifier

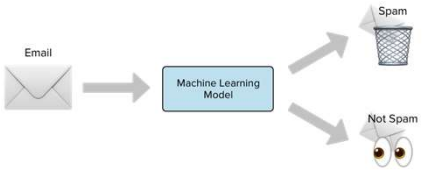


https://www.youtube.com/watch?v=O2L2Uv9pdDA

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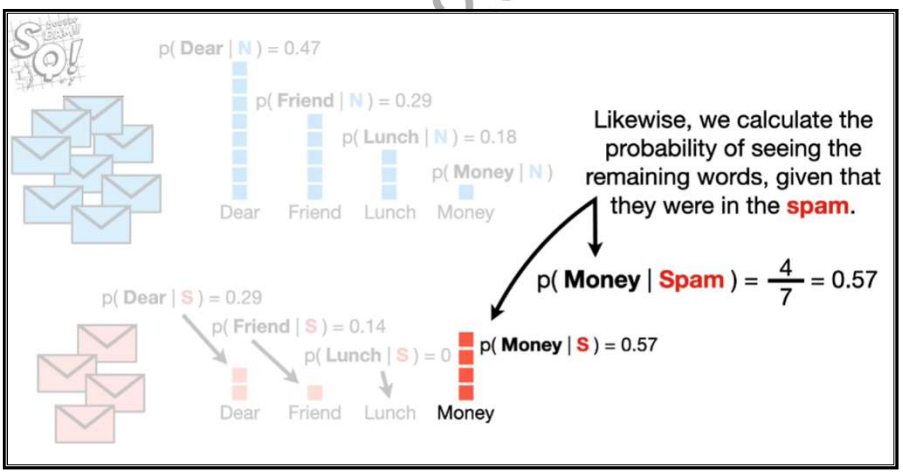
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Naive Bayes Classifier



The Problem:

- We want to know: Is this email Spam or Not Spam?
- We calculate which label has the highest probability.



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The Naive Independence Assumption

Naive Bayes assumes:

$$P(x_1, x_2, ..., x_n|y) = P(x_1|y) \cdot P(x_2|y) \cdots P(x_n|y)$$

$$P(y|x_1, ..., x_n) \propto P(y) \cdot \prod_{i=1}^{n} P(x_i|y)$$

Why "naive"? *(tính đơn giản hóa)*
It assumes features are independent, e.g.,

- The word occurrence of "free" is independent of "winner"
- The sender type is independent of email length

https://www.geeksforgeeks.org/machine-learning/naive-bayes-classifiers/

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Laplace Correction

- Zero-probability problem
  - What if a word NEVER appears in spam in training?
    Example if $P(\text{“congratulations”} \mid Spam) = 0$
    Then the entire posterior probability becomes 0
  - The zero-probability problem will cause the model is “corrupted”
- Solution: Laplace Correction (Smoothing)

$$P(X_i \mid Y) = \frac{count(X_i, Y) + 1}{count(Y) + k}$$

  - Where:
    k = number of possible feature values

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Example Features

Let’s classify emails using:

1. **Keyword presence:** contains “free” (yes/no)
2. **Sender type:** known / unknown
3. **Email length:** short / medium / long

These become features:

$$X = (x_1, x_2, x_3)$$

Our model must estimate:

$$P(x_1 \mid Spam)$$
$$P(x_2 \mid Spam)$$
$$P(x_3 \mid Spam)$$

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Lab 4.1: Spam Email – Data Setup

ID	contains free	sender	length	Label
1	yes	unknown	short	Spam
2	yes	known	short	Spam
3	no	unknown	short	Spam
4	yes	unknown	medium	Spam
5	no	unknown	long	Spam
6	no	known	medium	Spam
7	no	known	short	Not Spam
8	no	known	long	Not Spam
9	yes	known	medium	Not Spam
10	no	unknown	medium	Not Spam

Counts

- Total examples = **10**
- Spam = **6** examples
- Not Spam = **4** examples

$$P(Spam) = \frac{6}{10} = 0.6,$$

$$P(NotSpam) = \frac{4}{10} = 0.4$$

Features and their possible values

- x_1 :contains words "free" ∈ {yes, no} (k=2)
- x_2 :sender ∈ {known, unknown} (k=2)
- x_3 :length ∈ {short, medium, long} (k=3)

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Lab 4.1: Practices

ID	contains free	sender	length	Label
1	yes	unknown	short	Spam
2	yes	known	short	Spam
3	no	unknown	short	Spam
4	yes	unknown	medium	Spam
5	no	unknown	long	Spam
6	no	known	medium	Spam
7	no	known	short	Not Spam
8	no	known	long	Not Spam
9	yes	known	medium	Not Spam
10	no	unknown	medium	Not Spam

$$P(X_i \mid Y) = \frac{count(X_i, Y) + 1}{count(Y) + k}$$

Where:

- k= number of possible feature values
- avoids zero-probability problem

1. P(free=yes | Spam) = $\frac{3+1}{6+2} = \frac{4}{8} = 0.5$
2. P(free=no | Spam) = $\frac{3+1}{6+2} = \frac{4}{8} = 0.5$
3. P(sender=known | Spam) = ?
4. P(sender=unknown | Spam) = ?
5. P(length=short | Spam) = ?
6. P(length=medium | Spam) = ?
7. P(length=long | Spam) = ?

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Lab 4.1: Classify a new email

New email features:

- free = **yes**
- sender = **unknown**
- length = **short**



We compute each class:

$$P(X_1, X_2, \ldots, X_n \mid Y) = \prod_{i=1}^{n} P(X_i \mid Y)$$

We have X = (free = **yes,** sender = **unknown,** length = **short**)

P(X|Spam) = P(free = **yes** |Spam)
x P(sender = **unknown** |Spam)
x P(length = **short** |Spam)
= ?

P(X|Not Spam) = ?

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Naive Bayes — Key Takeaways

Pros

- Works well with text and high-dimensional data
- Handles noise surprisingly well
- Simple and interpretable probability model

Cons

- Assumes feature independence (often unrealistic)
- Zero probabilities require smoothing
- Lower accuracy when features interact strongly

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Decision Trees



https://www.youtube.com/watch?v=qbeewz1Z6ls

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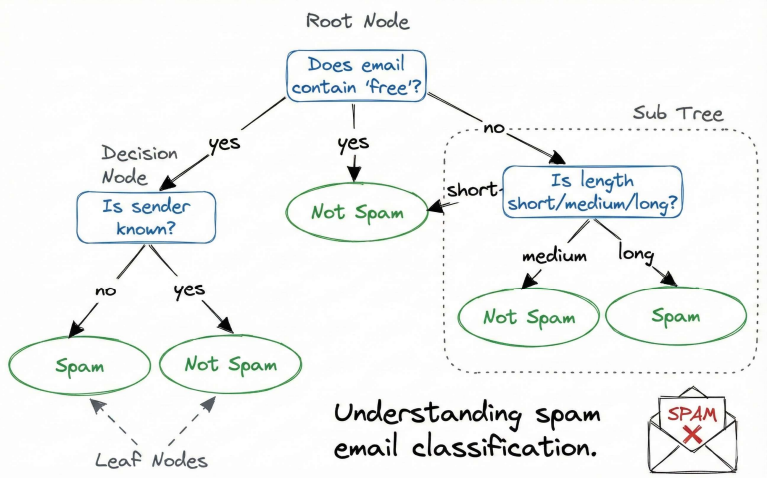
Decision Trees

Decision Trees answer the question:
"Which feature best splits the data into correct classes?"

They classify by repeatedly asking **simple questions** such as:

- *"Does email contain 'free'?"*
- *"Is sender known?"*
- *"Is length short/medium/long?"*

The result is a **tree** to help the decision, called **Decision Tree**

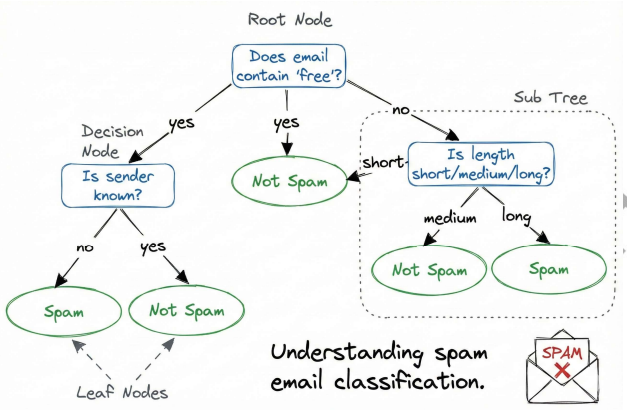


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Structure of a Decision Tree



Decision Tree consists of:

- **Root Node** — select the best features to start the splitting process
- **Internal Nodes** — questions
- **Branches** — answers (Boolean or categorical values)
- **Leaves** — class predictions (Spam / NotSpam)

Purity concepts

- **Purity level** – Number of leaves that the node contains
- **Mixed node** – contains leaves and other sub nodes
- **Pure node** – contains only leaves

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Lab 4:2 Spam email – Data Setup

We use a **small dataset** to construct a tree manually.
Total: 5 samples

- Spam = 3
- NotSpam = 2

ID	Free?	Sender	Length	Label
1	yes	unknown	short	Spam
2	yes	known	medium	Spam
3	no	unknown	short	Spam
4	no	known	medium	NotSpam
5	no	known	long	NotSpam

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Procedure

1. Check each feature's purity
2. Choose the feature with the most-pure node
3. Split the data
4. Repeat until all groups is pure

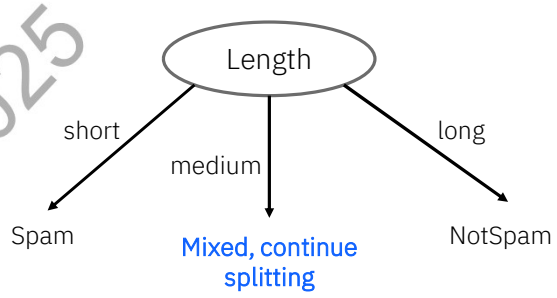
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Lab 4.2: Step 1&2 - Check each feature's purity

ID	Free?	Sender	Length	Label
1	yes	unknown	short	Spam
2	yes	known	medium	Spam
3	no	unknown	short	Spam
4	no	known	medium	NotSpam
5	no	known	long	NotSpam

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Split by feature Sender
P(unknown) = 2 Spam/ 0 NotSpam
P(known) = 1 Spam/ 2 NotSpam
➔ Purity level: 1

Split by feature Length
P(short) = 2 Spam/ 0 NotSpam
P(medium) = 1 Spam/ 1 NotSpam
P(long) = 0 Spam/ 1 NotSpam
➔ Purity level: 2

Select the feature that has the highest purity level
-> Select feature ***Length*** as the node of the tree (at the current level)

Length is a mixed node (still contains other sub-nodes, not only leaves)
-> The splitting process continues to the branch that it contains the sub node

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Lab 4.2: Step 3&4 - Split the data and Repeat

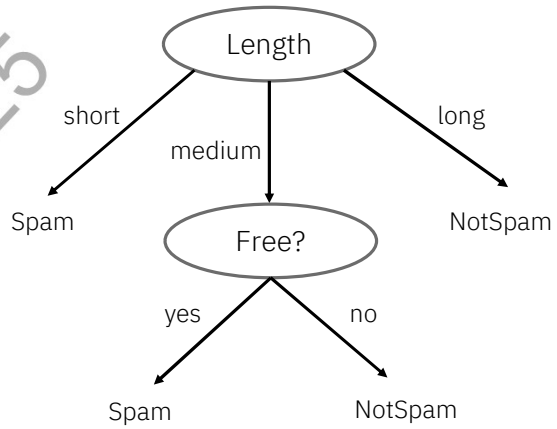
Only keep the "medium" length records

ID	Free?	Sender	Length	Label
~~1~~	~~yes~~	~~unknown~~	~~short~~	~~Spam~~
2	yes	known	medium	Spam
~~3~~	~~no~~	~~unknown~~	~~short~~	~~Spam~~
4	no	known	medium	NotSpam
~~5~~	~~no~~	~~known~~	~~long~~	~~NotSpam~~

Split by feature containing "Free"
P(yes) = 1 Spam/ 0 NotSpam
P(no) = 0 Spam/ 1 NotSpam
➔ Purity level: 2

The process stops as the selected node is pure
(not other sub-nodes, all are leaves)

Build the next level of the tree



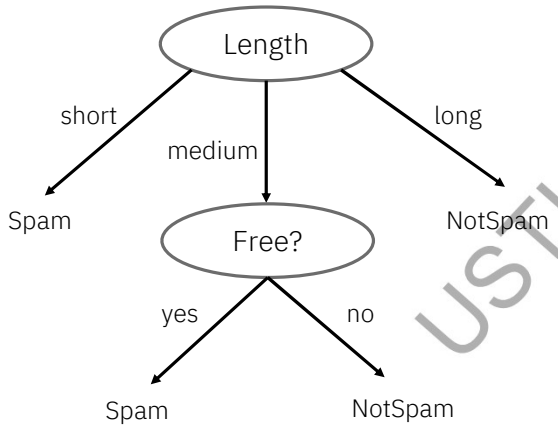
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Lab 4.2: Practice

With this decision tree



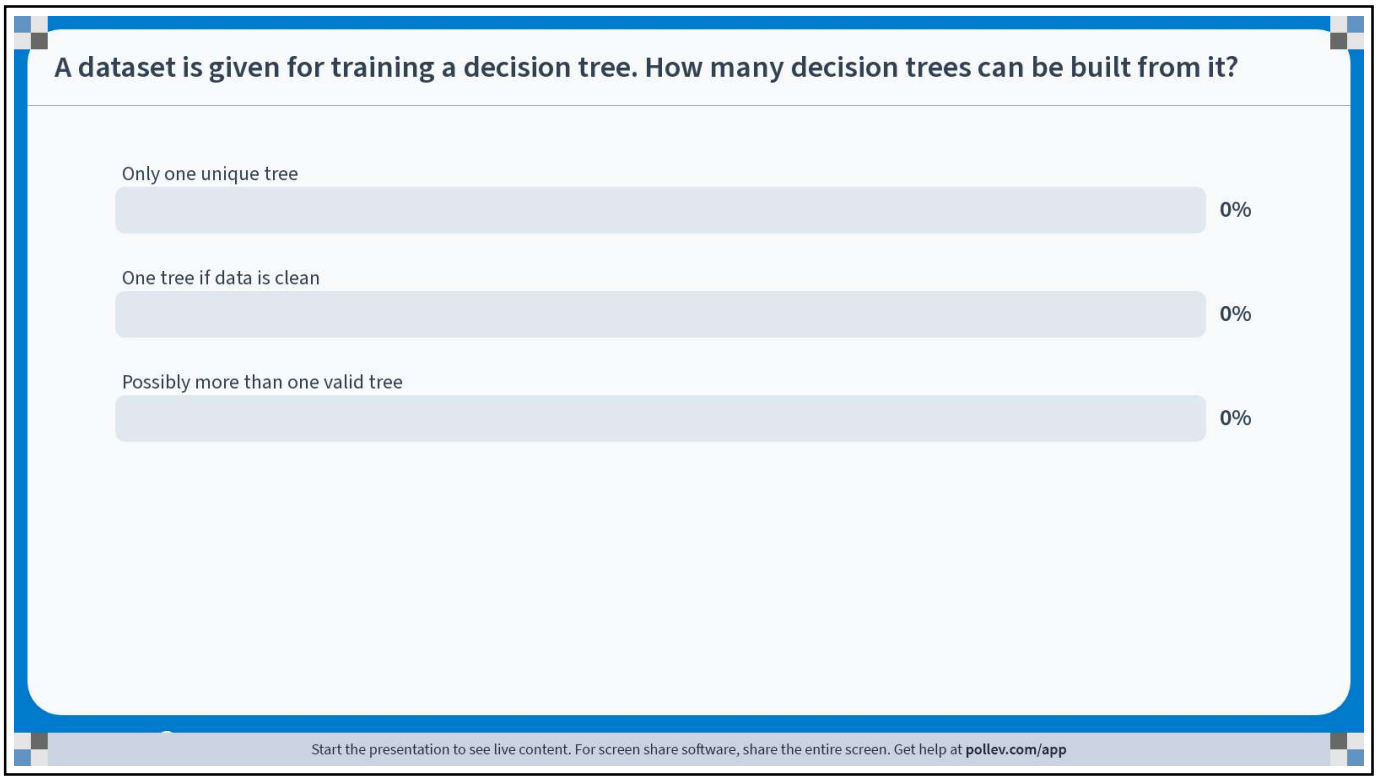
Now classify these new emails

Email 1: Free = no, Sender = unknown, Length = short
Email 2: Free = yes, Sender = known, Length = long
Email 3: Free = no, Sender = known, Length = medium
Email 4: Free = yes, Sender = unknown, Length = short
Email 5: Free = no, Sender = unknown, Length = long

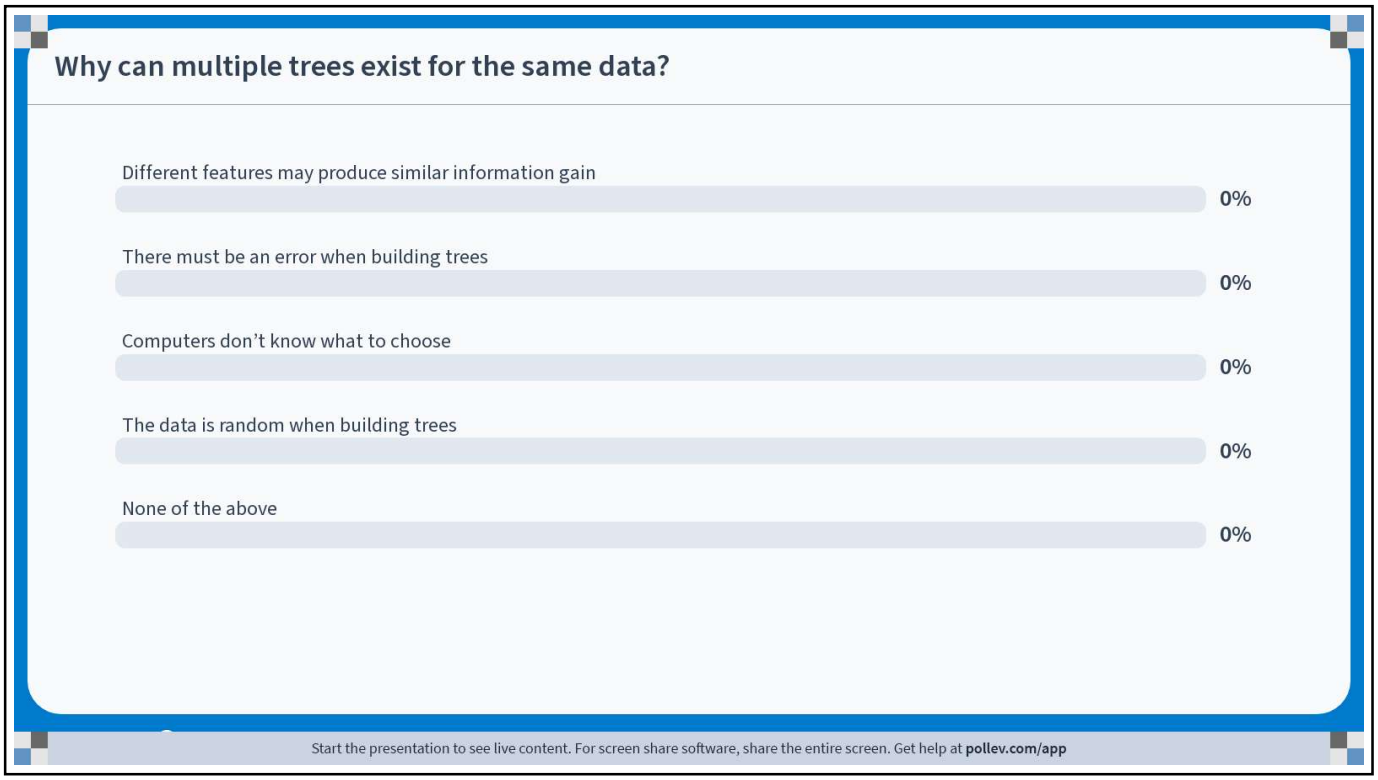
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A dataset is given for training a decision tree. How many decision trees can be built from it?
Only one unique tree
0%
One tree if data is clean
0%
Possibly more than one valid tree
0%
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Why can multiple trees exist for the same data?
Different features may produce similar information gain
0%
There must be an error when building trees
0%
Computers don't know what to choose
0%
The data is random when building trees
0%
None of the above
0%
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Decision Trees — Key Takeaways

Pros

- Highly interpretable — produces human-readable rules
- Handles categorical & numerical features
- No need for standardization or scaling

Cons

- Small data changes can reshape the tree (unstable)
- Can be shallow or biased if features dominate early splits
- Hard to scale well on high-dimensional text data compared to Naive Bayes

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Other tree-based ML models

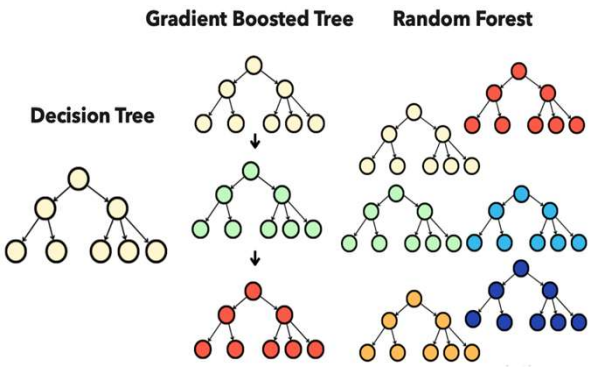
Advanced tree models improve Decision Trees primarily by addressing High Variance and Bias:

- Random Forests (RF): Reduces overfitting (variance) and instability.

  It builds multiple independent trees using random data/features (Bagging) and averages results for a robust, stable prediction.

- Gradient Boosting Machines (GBM): Improves accuracy (reduces bias).

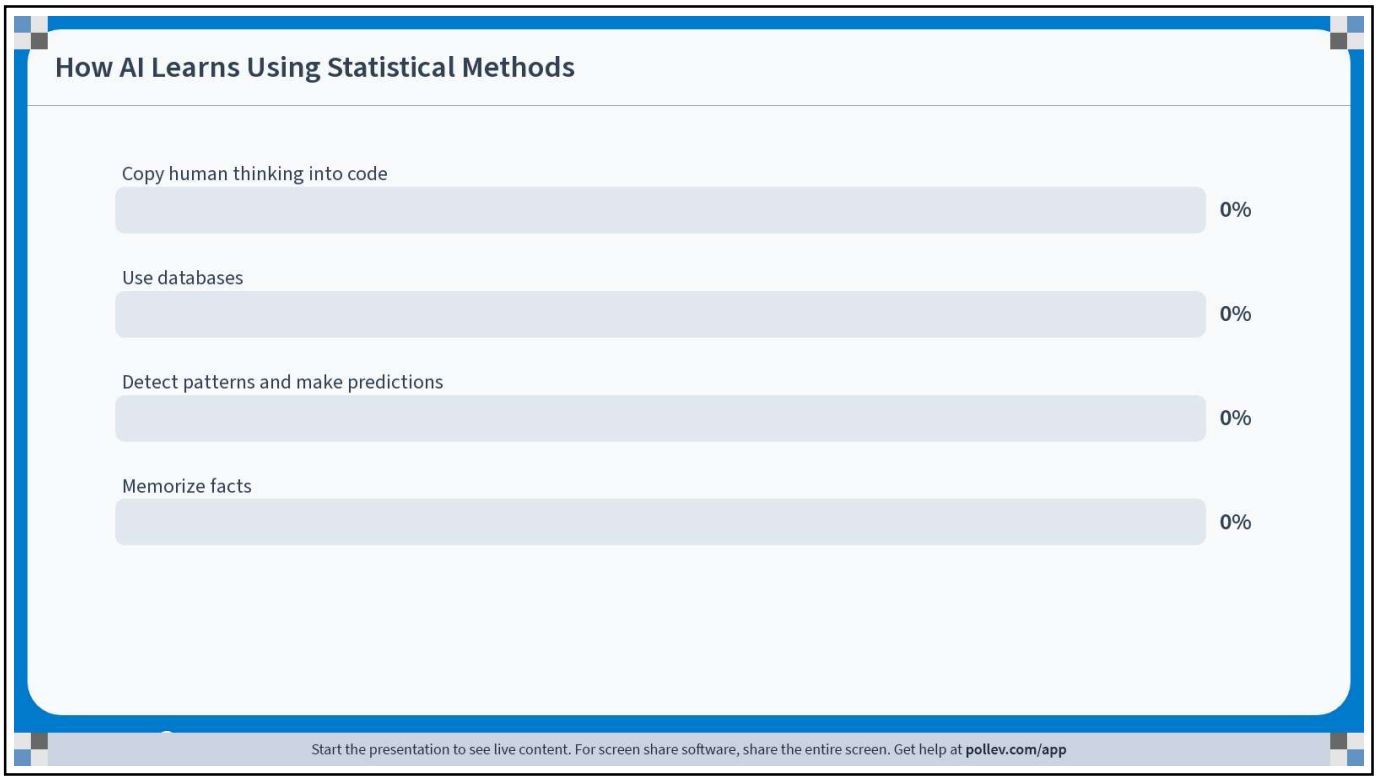
  It builds trees sequentially (Boosting), training each new tree to fix the prediction errors (residuals) of the previous ones, iteratively boosting performance.



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Thank you!

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